

WIRELESS TECHNOLOGY For Defence

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The defence industry is turning to a new generation of systems for faster, cheaper and more flexible communication, security and more

Communication is essential for military operations, although it is difficult to establish in many cases. Laying cables, deploying equipment and securing connections is time consuming, inflexible and limits users' mobility. Network deployment speed can be increased by using wireless technologies. These help military staff and commanders to stay connected and provide flexibility to move freely. However, a more connected world that integrates the Internet of Things (IoT), Cloud computing and advanced wireless communication also invites new kinds of threats. Network security is no longer just about protecting equipment and borders. Attacks now aim at security gaps in aircraft, sensors, satellites and space vehicles.

In military applications vast amounts of data flow through sensors. This requires a single data stream that can combine video feeds from drones targeting radars and mobile phone intercepts. With reduced weight, size and power consumption, it has become easier to integrate more sensors into soldier systems. Within the defence world, information needs to be communicated in a wireless way, from sensors to war fighters.

Nitin Shetty, chief executive officer, Convergent Technologies, says, "We provide two solutions for wireless communication for defence: local and remote spec-

trum monitoring and surveillance systems, and Global Positioning System (GPS) protection against spoofing and jamming threats."

He adds that, assured access and awareness of radio frequency (RF) spectrum are critical for a wide range of military and security applications. There are critical national infrastructures that rely on GPS-delivered positions, navigation and timing for daily operations. There is a need to protect these installations.

Convergent Technologies provides solutions for regulatory monitoring and management, finding direction and geolocation of emitters, technical surveillance countermeasures, remote sites, and perimeter and border monitoring, among others. They provide protection against the vulnerability of GPS systems to various signal incidents. Their solutions are better and secure as they provide full turnkey spectrum monitoring and management solutions and in-country support and service.

They provide indoor, outdoor, portable and IP67-compliant solutions that are based on a programmable anomaly detector that validates GPS sub-frames for spoofing incidents. Further, new validation rules are dynamically loaded as new threats are identified. There is also input capability from an atomic clock reference that enables users to enhance the holdover performance for accurate time.

Girish Baliga, marketing manager, Keysight Technologies India Pvt Ltd, says, "Keysight is a leading solution provider to the aerospace and defence sector for Internet, wireless communication, satellites, space and navigation, radar, electronic warfare systems and signal intelligence. Radar and satellite communication systems require higher frequencies, larger bandwidths and greater sensitivity to support uninterrupted connections."

In network-centric warfare, both speed of information sharing and quality of information form an integral part of the information domain. This is where the choice of



Fig. 1: Integrated sensors in a soldier system (Credit: www.analog.com)